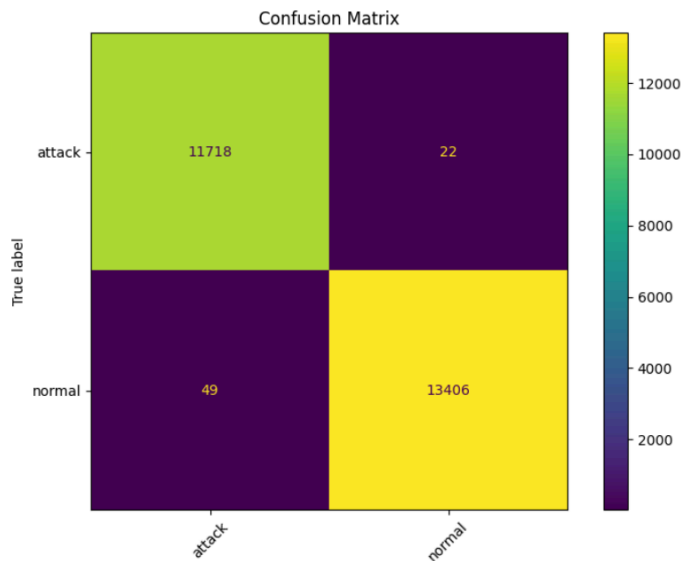


3. [Pro] (Implementing a DNN on the dataset, 5 pts x 3 tasks = 15 pts)  
Complete the following tasks in the provided Python notebook (.ipynb) file:
  - (a) Implement a **deep neural network (DNN)** on the **Network Anomaly Dataset**. Ensure to include **two neurons** and **softmax activation** in the output layer of your DNN.
  - (b) Compile and train your DNN model on the **training set (X\_train)**. Denote the trained model as **base\_model**.
  - (c) Evaluate the **base\_model** on the **test set (X\_test)** using **classification\_report** and **confusion\_matrix** from the **sklearn.metrics** library. **Report these numbers in your .pdf writeup file using screenshots.**
4. [Pro] (Implementing Quantized Model, 10 pts x 2 tasks = 20 pts)  
Complete the following tasks in the provided Python notebook (.ipynb) file:
  - (a) Implement **Dynamic Range Quantization** on the **base\_model**. Designate the resulting quantized ML model as **tfite\_quant\_model**.
  - (b) Evaluate the **tfite\_quant\_model** on the **test set (X\_test)** using **classification\_report** and **confusion\_matrix** from the **sklearn.metrics** library. **Report these numbers in your .pdf writeup file using screenshots.**

3)

Test Accuracy: 0.9972

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| attack       | 1.00      | 1.00   | 1.00     | 11740   |
| normal       | 1.00      | 1.00   | 1.00     | 13455   |
| accuracy     |           |        | 1.00     | 25195   |
| macro avg    | 1.00      | 1.00   | 1.00     | 25195   |
| weighted avg | 1.00      | 1.00   | 1.00     | 25195   |



4)

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| normal       | 1.00      | 1.00   | 1.00     | 11740   |
| attack       | 1.00      | 1.00   | 1.00     | 13455   |
| accuracy     |           |        | 1.00     | 25195   |
| macro avg    | 1.00      | 1.00   | 1.00     | 25195   |
| weighted avg | 1.00      | 1.00   | 1.00     | 25195   |

